StepperTune

User Manual

MIDI to G-code music engine

Guide to melody creation, MIDI import, jingle generation and installation in Bambu Studio.

1. Introduction

StepperTune converts manually entered melodies or imported MIDI files into G-code for start and end-print jingles on supported Bambu Lab printers. The application provides a PC preview and automatically applies the correct driver for the selected printer model.

> **Important:** StepperTune generates the musical section to be inserted into the printer preset. It does not replace normal print G-code.

2. Interface

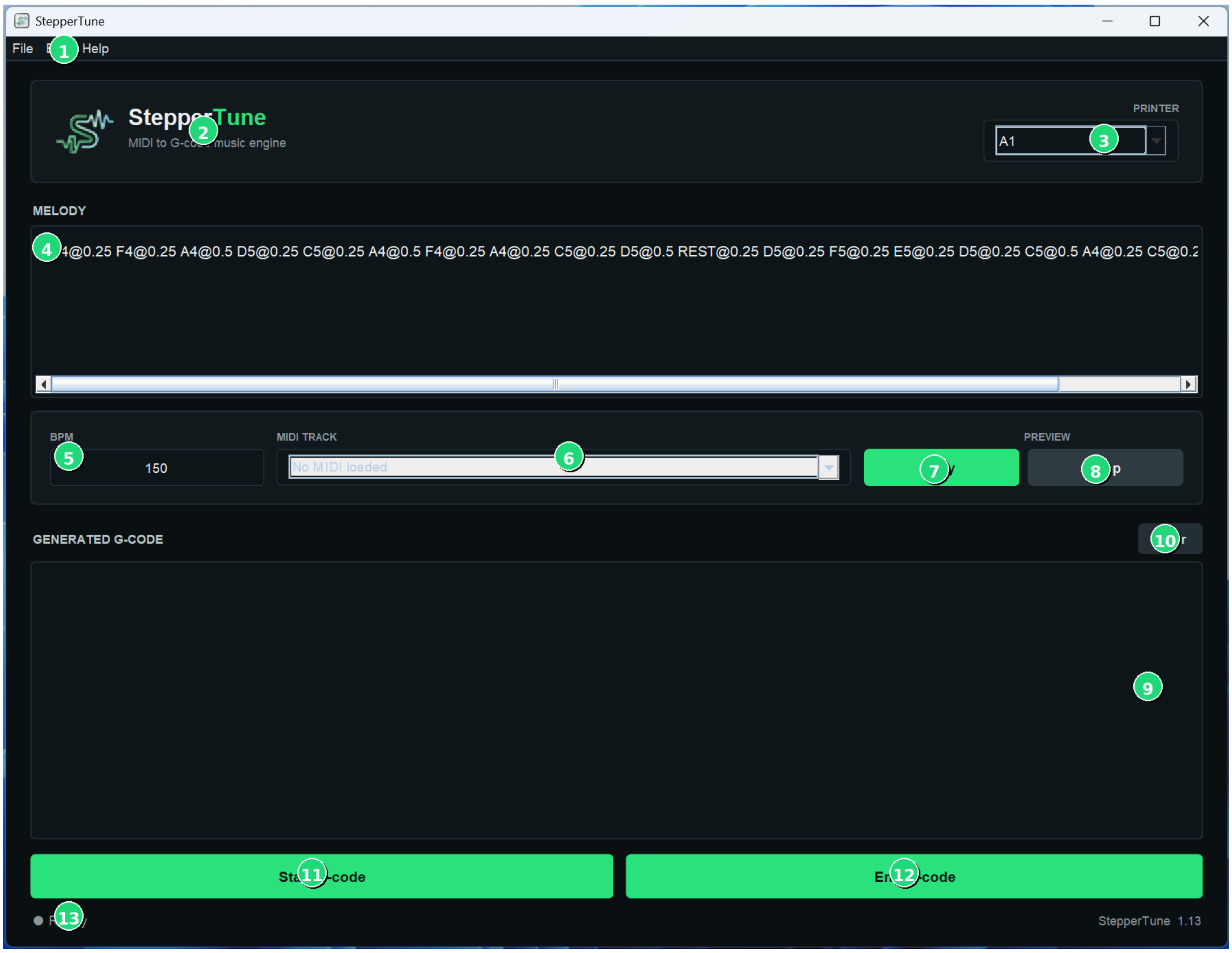

Figure 1 - Main interface with numbered references.

Ref.	Control	Function
1	Main menu	Access to File, Edit and Help.
2	Logo and identification	Application name and description.
3	Printer	Select the printer model.
4	Melody	Enter or display the melody.

5	BPM	Melody tempo in beats per minute.
6	MIDI Track	Select the imported MIDI track.
7	Play	Plays the preview.
8	Stop	Stops the preview.
9	Generated G-code	Displays generated G-code.
10	Clear	Clears the G-code output.
11	Start G-code	Generates the start-print jingle.
12	End G-code	Generates the end-print jingle.
13	Status bar	Shows status and operation confirmations.

3. Creating a melody

Notes can be typed directly into the Melody area. Each event uses the Note@Duration format and events are separated by spaces.

C4@0.25 D4@0.25 E4@0.5 REST@0.25 G4@0.5

3.1 Notes, octaves and rests

- Available notes: C, C#, D, D#, E, F, F#, G, G#, A, A#, B.
- The number after the note indicates the octave, for example C4 or F#5.
- REST inserts a pause and uses the same duration syntax, for example REST@0.25.

3.2 Duration and BPM

Values such as 1, 0.5, 0.25 and 0.125 represent relative musical durations. BPM determines the actual sequence speed.

4. Importing a MIDI file

1. Open File > Load MIDI and select a .mid file.
2. If the MIDI contains multiple tracks, select the desired one in MIDI Track.
3. Check the converted melody and BPM.
4. Press Play to preview the melody before generation.

With complex MIDI files, track selection can significantly affect the result. Tracks with a clear, less-overlapped melodic line usually work best.

5. Preview and generation

5. Select the correct model in Printer.
6. Prepare the melody manually or via MIDI.
7. Use Play and Stop to check the preview.
8. Press Start G-code for the start jingle or End G-code for the end-print jingle.
9. The result appears in Generated G-code and can be exported or copied.

Note: The preview uses the computer audio system. The printer's actual timbre is produced by motor vibration and will therefore sound different.

6. Adding the jingle in Bambu Studio

The following procedure is adapted from Bambu Lab's official MIDI/G-code sound documentation.

Bambu Lab Wiki - MIDI

6.1 Select printer and preset

In Bambu Studio, on the Prepare tab, select the printer you are using and the correct nozzle size. Then click Edit preset next to the printer name.

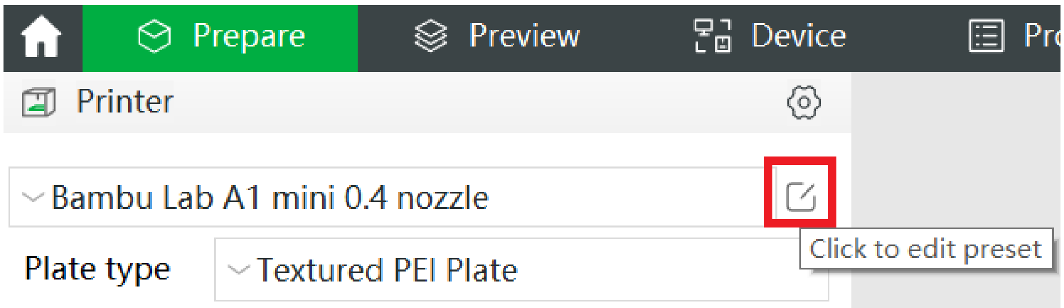

Figure 2 - Opening the printer preset. Image source: Bambu Lab Wiki.

6.2 Enable Advanced and open Machine G-code

In the Printer settings window, enable Advanced at the top right, then open the Machine G-code tab.

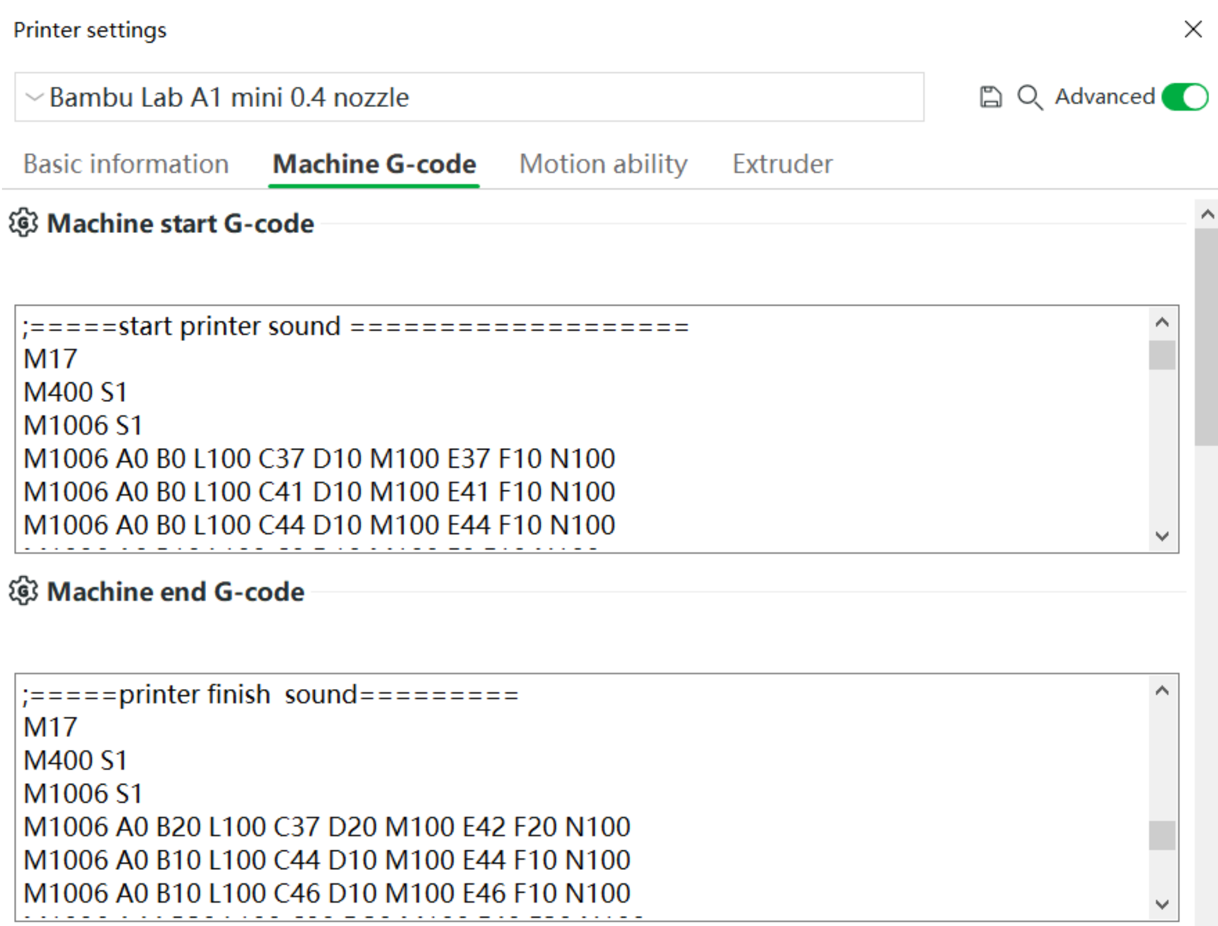

Figure 3 - Machine G-code tab. Image source: Bambu Lab Wiki.

6.3 Replace only the sound section

For the start sound, in Machine start G-code locate the divider:

start printer sound

Replace only the code between this divider and the next one with the code generated by Start G-code.

For the end sound, in Machine end G-code locate the divider:

printer finish sound

Replace only the corresponding sound block with the code generated by End G-code.

WARNING: Do not replace the entire Machine start G-code or Machine end G-code. Modify only the sound block and leave all other machine G-code unchanged.

Line breaks: When copying and pasting, preserve the generated G-code line breaks exactly. Bambu Lab documentation warns that lost line breaks can cause G-code errors.

6.4 Save the preset

Click Save. You can save the configuration as a User Preset, making it available to all projects using that preset, or as a Project Inside Preset, limiting it to the current project.

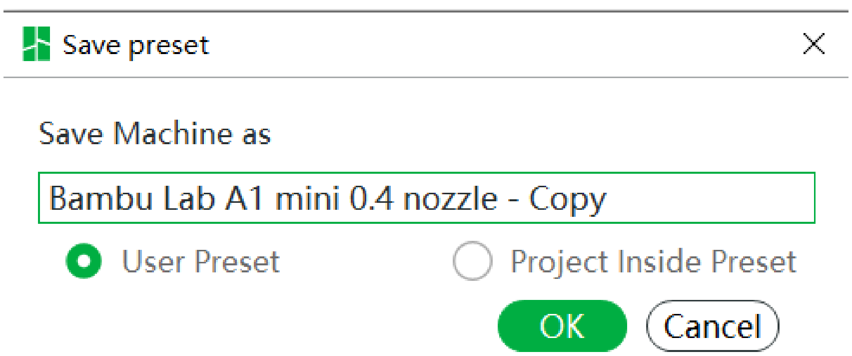

Figure 4 - Saving the preset. Image source: Bambu Lab Wiki.

7. Supported printers

- A1
- A1 mini
- H2S
- H2D
- X2D
- P2S

Always select the model actually being used. StepperTune automatically applies the sound parameters and G-code structure defined by the corresponding driver.

8. Usage tips

- Start with short, recognizable jingles.
- Always preview the melody before generating G-code.
- With complex MIDI files, try multiple tracks and choose the cleanest one.
- Keep a copy of the original printer preset before modifying it.
- After a change, initially perform a controlled print or test.

9. Responsibility and safety

StepperTune is an independent project and is not an official Bambu Lab product. It is not affiliated with, sponsored by, or endorsed by Bambu Lab.

The software generates G-code commands that operate the printer motors to produce sound. The user is responsible for verifying the selected model, the generated content, and its correct insertion into the printer preset.

Use of manually modified G-code, use on unsupported models, or replacement of Machine G-code sections other than the dedicated sound block is entirely at the user's own responsibility.

Firmware, Bambu Studio and G-code command behavior may change over time. Always verify correct operation after printer firmware or software updates.

The StepperTune author cannot be held responsible for printer damage, failed prints, data loss, or other effects resulting from improper use, manual modifications, or use on unsupported hardware.

10. References

Bambu Studio procedure and reference images: Bambu Lab Wiki - MIDI

Bambu Lab, Bambu Studio and the referenced model names belong to their respective owners.